

Bhanu Saisree Thanniru

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SUMMARY

Software Engineer with M.S. in Computer Science, specializing in building responsive and scalable web applications and distributed services. Proficient in developing microservices with Java, Spring Boot, and Angular, React, integrating REST APIs, debugging complex issues, and implementing robust CI/CD pipelines using Jenkins and GitHub Actions. Strong problem-solver with a focus on user-centric solutions, performance optimization, and high code quality. Self-starter, collaborative team member, and active user of GenAI tools (Copilot, Cursor, ChatGPT) and Azure AI Foundry for faster, higher-quality delivery.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C

Backend / Services: Spring (Core, Boot, MVC, Data JPA, Security), Java/J2EE, Node.js, Express.js, REST APIs, GraphQL, SOAP APIs, Microservices, Distributed Systems

Frontend: React, Angular17, HTML5, CSS3, JavaScript, TypeScript, Material-UI, Redux

Databases: Azure SQL, SQL Server, PostgreSQL, MySQL, MongoDB, Oracle

Cloud / DevOps: Microsoft Azure (App Service, Azure SQL, Azure DevOps, Azure AI Foundry), AWS (EC2, S3, RDS, Lambda, EKS, Fargate, CloudFormation), Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions, Azure Pipelines)

Testing / Tools: JUnit, Mockito, Jest, Cypress, Jasmine, Postman, SOAP UI, Git, GitHub, SVN, Azure DevOps, Swagger/OpenAPI, Observability (logging, metrics), Tableau, Joget DX

AI / LLM: Anthropic Claude API, Claude Code, Prompt Engineering, LLM Integration, Structured Output Extraction

EXPERIENCE

Software Engineer

Northern Kytes

Feb 2026 – May 2026

Remote

- Engineered an LLM-powered document extraction pipeline using Anthropic Claude API and prompt engineering to parse unstructured documents into structured fields, persisted via SQLAlchemy to Azure SQL.
- Built a scalable multi-tenant SaaS platform using React, FastAPI, and Python, building secure RESTful services for authentication, document lifecycle management, and tenant administration.
- Leveraged Generative AI tools (Cursor, Claude) to streamline development, reduce implementation time, and accelerate feature delivery.
- Implemented JWT authentication and RBAC across three privilege tiers (User / Admin / Super Admin) with aligned authorization logic across React frontend and FastAPI backend.
- Designed SQLAlchemy data models with Azure SQL and integrated Azure Blob Storage for document ingestion, ensuring multi-tenant data isolation and enterprise-grade security.
- Developed reusable React components with react-hook-form and Yup validation, and supported containerized deployments using Docker and Azure Container Apps.

Graduate Assistant

University of North Carolina at Charlotte

Mar 2024 – Dec 2025

Charlotte, NC

- Built full-stack features for internal research and departmental tools using Java 17/Spring Boot REST APIs and Angular 17, improving how faculty and students collect, query, and visualize research data.
- Created SQL queries and stored procedures in PostgreSQL and Azure SQL to generate and optimize reports for faculty.
- Developed Python scripts for data processing and ML experiments using NumPy, Pandas, and Scikit-Learn to support research projects.
- Worked with professors in Agile-style sprints using Git, code reviews, and iterative feedback, demonstrating a growth mindset and strong collaboration skills.

Senior Software Engineer

Cappemini

Sep 2022 – Jan 2024

India

- Developed and maintained scalable full-stack web applications using Java/Spring Boot microservices for backend and React.js for frontend, delivering high-performance banking applications serving millions of transactions daily.
- Led migration initiatives transforming legacy monolithic systems to microservices architecture on AWS, improving application performance by ~40%, and enhancing system maintainability and scalability.
- Architected and implemented RESTful APIs and event-driven microservices using Spring Boot, Spring Data JPA, Kafka, and Oracle/PostgreSQL, ensuring low latency, and fault tolerance.
- Optimized frontend performance by refactoring React components and implementing efficient state management with Redux, lazy loading, and code splitting to reduce initial load time by 30%.
- Implemented a comprehensive testing strategy using JUnit and Mockito for unit and integration tests, achieving around 95% code coverage and significantly reducing production defects.
- Automated cloud infrastructure provisioning and deployment using AWS CloudFormation, EC2 auto-scaling, and EKS for container orchestration, integrating CI/CD pipelines with Jenkins and GitHub Actions to reduce release cycle time by 60%.
- Modeled workflows in BPM tool and integrated them with Java/Spring back-end endpoints for configuration-driven changes in form and workflow behavior.
- Collaborated cross-functionally with product managers, QA engineers, and operations teams in Agile/Scrum ceremonies and conducted code reviews to maintain high code quality and design consistency.
- Documented technical architecture, API contracts (Swagger), and service dependencies in Confluence to improve knowledge sharing and onboarding efficiency across distributed teams.

Software Engineer
Capgemini

Feb 2021 – Aug 2022
India

- Developed enterprise web applications using Java/J2EE, Spring MVC, Hibernate, JDBC, and relational databases (MySQL, PostgreSQL, Oracle) to implement core business logic and data access layers for banking workflows.
- Built responsive and interactive frontend features using Angular, HTML5, CSS3, JavaScript, and TypeScript, ensuring cross-browser compatibility and seamless user experience across desktop and mobile devices.
- Created RESTful web services using Spring Boot to enable integration between internal systems and third-party services, implementing proper error handling, input validation, and API versioning strategies.
- Enhanced application security by implementing Spring Security with JWT-based role-based access control (RBAC), standardizing authentication and authorization patterns across microservices and reducing security vulnerabilities.
- Engineered BPM-based workflow solutions consuming SOAP APIs, implemented backend data storage, and built reporting dashboards for operational visibility.
- Utilized version control systems (Git, SVN) for collaborative development, maintained clean branching strategies (feature branches, pull requests), and participated in code reviews to ensure code quality and knowledge transfer.
- Worked in Agile teams following Scrum methodology, actively contributing to sprint planning, daily standups, backlog grooming, retrospectives, and iterative delivery of features aligned with business priorities.

PROJECTS

Real-time Wav2Lip Application

PyTorch, OpenVINO, Streamlit, CUDA

- Built a GPU-optimized lip-sync system using PyTorch, Generative Adversarial Networks (GANs), SyncNet, ResNet, and CNN encoder-decoder architectures with mel-spectrogram-driven audio-video synchronization on the LRS2 dataset.
- Implemented audio-video preprocessing pipelines with Librosa, FFmpeg, and OpenCV, and trained a custom discriminator using cosine similarity + BCE loss to improve lip-sync quality and temporal alignment.
- Designed a real-time inference pipeline using CUDA, NumPy, Pandas, and Scikit-Learn, and exposed it via a Streamlit web interface for rapid experimentation and interactive demos.

Online Pizza Ordering Application

Spring Boot, Angular, Oracle

- Developed a microservice-based e-commerce web application with Spring Boot backend services and an Angular frontend to support role-based access control for administrators and customers with distinct user workflows.
- Implemented RESTful microservices for menu management, order creation, shopping cart functionality, and payment processing, using Oracle as the transactional data store for orders and inventory.

Personal Budget Planning Application

Node.js, Angular 17, Firebase

- Built a full-stack budget management tool to track income, expenses, and financial goals through an interactive Angular dashboard with real-time data synchronization and responsive UI components.
- Implemented secure user authentication with Firebase Auth and created dynamic data visualizations using Chart.js for spending trends, category breakdowns, and monthly budget tracking.

EDUCATION

University of North Carolina at Charlotte

Dec 2025

Master of Science in Computer Science — CGPA: 4.0

Relevant Coursework: Software System Design and Implementation, Data Structures, Network Based Application Development, Cloud Computing for Data Analysis

Vignan's Lara Institute of Technology and Science

Sept 2020

Bachelor of Technology in Information Technology — CGPA: 3.52

Relevant Coursework: Java Programming, Python, C, Database Management System, Web Technologies, Operating Systems, Unix and Shell Scripting